PCA

In [ ]:
```
100 columns  --->2,3 Columns  -->PCA
```

In [ ]:
```
Student     Math  Pysics  Chem   English

A           85     88      82      72
B           90     92      89      75
C           60     65      62      80

PCA  Instead of Keeping all three seperatly it can create a new component

PCA1=Combination of math+phy+chem+eng

Original 4 --->2 principal component

Dimensionality Reduction

X1
X2
X3  --->PCA -->PCA1,PCA2,PCA3
X4
X5



Dataset

Standardize

Covariance matrix

Eigen values & Vectors
Eigen values tell us how much variance is captured by each PCA

Example:

PC1 Eg value =4.5
PC2 Eg value =2.0
PC3 Eg value =0.8
PC4 Eg value =0.2
PC4 Eg value =0.2

Total variance=7.5

PC1=4.5/7.5=60%



Select PCA
Transform data
Reduce dataset
```

In [2]:
```
from sklearn.datasets import load_iris
import pandas as pd
```

In [4]:
```
iris=load_iris()
```

In [16]:
```
X=iris.data
Y=iris.target
```

In [28]:
```
df=pd.DataFrame(
    X,
    columns=iris.feature_names
)
```

In [30]:
```
df
```

Out[30]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2
...	...	...	...	...
145	6.7	3.0	5.2	2.3
146	6.3	2.5	5.0	1.9
147	6.5	3.0	5.2	2.0

```
PC1=4.5/7.5=60%



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In [30]:
```
df
```

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3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2
...	...	...	...	...
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146	6.3	2.5	5.0	1.9
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148	6.2	3.4	5.4	2.3

```
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4	5.0	3.6	1.4	0.2
147	6.5	3.0	5.2	2.0
148	6.2	3.4	5.4	2.3
149	5.9	3.0	5.1	1.8

150 rows × 4 columns

In [32]:
```
df.shape
```

Out[32]:
```
(150, 4)
```

In [34]:
```
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()

scaler=StandardScaler()
x_scaled=scaler.fit_transform(X)
```

In [36]:
```
x_scaled
```

Out[36]:
```
array([[-9.00681170e-01,  1.01900435e+00, -1.34022653e+00,
        -1.31544430e+00],
       [-1.14301691e+00, -1.31979479e-01, -1.34022653e+00,
        -1.31544430e+00],
       [-1.38535265e+00,  3.28414053e-01, -1.39706395e+00,
        -1.31544430e+00],
       [-1.50652052e+00,  9.82172869e-02, -1.28338910e+00,
        -1.31544430e+00],
       [-1.02184904e+00,  1.24920112e+00, -1.34022653e+00,
        -1.31544430e+00],
       [-5.37177559e-01,  1.93979142e+00, -1.16971425e+00,
        -1.05217993e+00],
       [-1.50652052e+00,  7.88807586e-01, -1.34022653e+00,
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        -1.31544430e+00],
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        -1.31544430e+00],
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        -1.44707648e+00],
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        -1.31544430e+00],
       [-1.26418478e+00,  7.88807586e-01, -1.22655167e+00,
        -1.31544430e+00],
       [-1.26418478e+00, -1.31979479e-01, -1.34022653e+00,
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        -1.44707648e+00],
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        -1.31544430e+00],
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        -1.05217993e+00],
       [-5.37177559e-01,  1.93979142e+00, -1.39706395e+00,
        -1.05217993e+00],
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        -1.31544430e+00],
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        -1.44707648e+00],
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        -1.05217993e+00],
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        -1.31544430e+00],
       [-1.50652052e+00,  3.28414053e-01, -1.34022653e+00,
        -1.31544430e+00],
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         1.32509732e-01],
       [-7.79513300e-01, -8.22569778e-01,  8.07091462e-02,
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        -2.62386821e-01],
       [ 6.86617933e-02, -1.31979479e-01,  2.51221427e-01,
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         2.64141916e-01],
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         2.64141916e-01],
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         3.95774101e-01],
       [-5.25060772e-02, -8.22569778e-01,  1.94384000e-01,
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       [-2.94841818e-01, -1.28296331e+00,  8.07091462e-02,
        -1.30754636e-01],
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         1.32509732e-01],
       [ 5.53333275e-01, -1.28296331e+00,  6.49083415e-01,
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       [ 6.74501145e-01, -3.62176246e-01,  3.08058854e-01,
         1.32509732e-01],
       [ 9.16836886e-01, -1.31979479e-01,  3.64896281e-01,
         2.64141916e-01],
       [ 1.15917263e+00, -5.92373012e-01,  5.92245988e-01,
         2.64141916e-01],
       [ 1.03800476e+00, -1.31979479e-01,  7.05920842e-01,
         6.59038469e-01],
       [ 1.89829664e-01, -3.62176246e-01,  4.21733708e-01,
         3.95774101e-01],
       [-1.73673948e-01, -1.05276654e+00, -1.46640561e-01,
       [ 1.89829664e-01,  7.88807586e-01,  4.21733708e-01,
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       [-2.94841818e-01, -1.31979479e-01,  1.94384000e-01,
         1.32509732e-01],
       [-4.16009689e-01, -1.28296331e+00,  1.37546573e-01,
         1.32509732e-01],
       [-4.16009689e-01, -1.05276654e+00,  3.64896281e-01,
         8.77547895e-04],
       [ 3.10997534e-01, -1.31979479e-01,  4.78571135e-01,
         2.64141916e-01],
       [-5.25060772e-02, -1.05276654e+00,  1.37546573e-01,
         8.77547895e-04],
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        -2.62386821e-01],
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         1.32509732e-01],
       [-1.73673948e-01, -1.31979479e-01,  2.51221427e-01,
         8.77547895e-04],
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         1.32509732e-01],
       [ 4.32165405e-01, -3.62176246e-01,  3.08058854e-01,
         1.32509732e-01],
       [-9.00681170e-01, -1.28296331e+00, -4.30827696e-01,
        -1.30754636e-01],
       [-1.73673948e-01, -5.92373012e-01,  1.94384000e-01,
         1.32509732e-01],
       [ 5.53333275e-01,  5.58610819e-01,  1.27429511e+00,
         1.71209594e+00],
       [-5.25060772e-02, -8.22569778e-01,  7.62758269e-01,
         9.22302838e-01],
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       [ 5.53333275e-01, -3.62176246e-01,  1.04694540e+00,
         7.90670654e-01],
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         1.31719939e+00],
       [ 2.12851559e+00, -1.31979479e-01,  1.61531967e+00,
         1.18556721e+00],
       [-1.14301691e+00, -1.28296331e+00,  4.21733708e-01,
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```

0	-2.264703	0.480027
1	-2.080961	-0.674134
2	-2.364229	-0.341908
3	-2.299384	-0.597395
4	-2.389842	0.646835
...	...	...
145	1.870503	0.386966
146	1.564580	-0.896687
147	1.521170	0.269069
148	1.372788	1.011254
149	0.960656	-0.024332